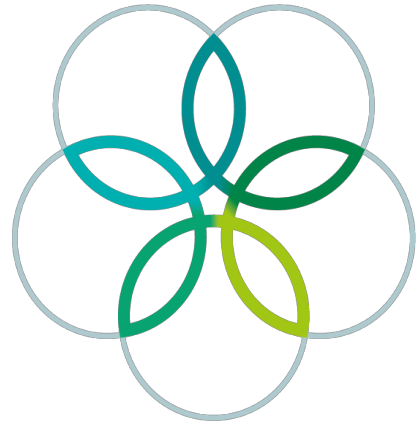
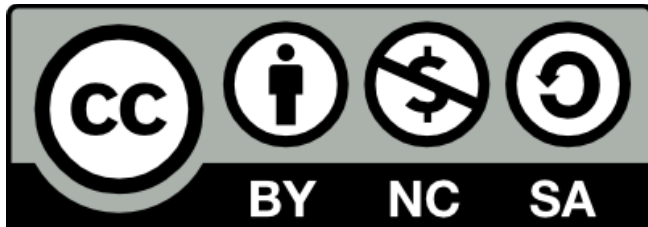


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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8th INTERNATIONAL BIOLOGY OLYMPIAD
PRACTICAL TEST-A
BOTANY AND BOTANIC ECOLOGY

1. **MATERIAL NUMBER: 1**
The function of this organism is.....
A) to fix free nitrogen
B) to convert proteins to NH_3
C) to make alcoholic fermentation
D) to make lactic acid fermentation
E) to produce amino acid from NH_3

2. **MATERIAL NUMBER: 1**
This organism is a (n).....
A) Eucaryot
B) Cyanobacteria
C) Yellow - green algae
D) Diatom
E) Green algae

3. **MATERIAL NUMBER: 2**
This organism is a(n).....
A) Procaryot
B) Metazoan
C) Diatom
D) Cyanobacteria
E) Green algae

4. **MATERIAL NUMBER: 2**
This organism
A) produces NO_2 from NH_3
B) produces NO_3 from NO_2
C) produces free nitrogen
D) forms the silica layers of the earth
E) is a zooplankton

5. **MATERIAL NUMBER: 2**
The species of this organism is
A) Anabaena sp.
B) Escherichia sp.
C) Navicula sp.
D) Chlorella sp.
E) Sargassum sp.

6. **MATERIAL NUMBER: 3**
It makes.....
A) Chemosynthesis
B) Photosynthesis
C) Nitrification
D) Denitrification
E) Fermentation

7. **MATERIAL NUMBER: 4**
Type of reproduction is.....
A) Autogamy

- B) Isogamous
- C) Anisogamous
- D) Conjugation
- E) Vegetative

8. MATERIAL NUMBER: 4

The shape of chloroplast is.....

- A) Star
- B) Oval
- C) Spiral
- D) Rod
- E) Circle

9. MATERIAL NUMBER: 5

The shape of bacteria is.....

- A) Spirillum
- B) Bacillus
- C) Coccus
- D) Vibrio
- E) Staphylococcus

10. MATERIAL NUMBER: 5

The group of bacteria is.....

- A) Gram (+)
- B) Gram (-)
- C) Gram (+) and Gram (-)
- D) One flagellated
- E) Poly flagellated

11. MATERIAL NUMBER: 6

The shape of bacteria is.....

- A) Spirillum
- B) Bacillus
- C) Coccus
- D) Vibrio
- E) Staphylococcus

12. MATERIAL NUMBER: 6

The group of bacteria is.....

- A) Gram (+)
- B) Gram (-)
- C) Gram (+) and Gram (-)
- D) One flagellated
- E) Many flagellated

13. MATERIAL NUMBER: 7

The number of flagella on the bacteria is.....

- A) 0
- B) 1
- C) 2
- D) 4
- E) Many

14. MATERIAL NUMBER: 8

The shape of bacteria is.....

- A) Staphylococcus
- B) Diplococcus
- C) Streptococcus
- D) Micrococcus
- E) Spirillum

15. MATERIAL NUMBER: 9

The shape of bacteria is.....

- A) Staphylococcus
- B) Diplococcus
- C) Streptococcus
- D) Micrococcus
- E) Spirillum

16. MATERIAL NUMBER: 10

Which of the followings includes this organism?

- A) Zygomycota
- B) Ascomycete
- C) Basidiomycete
- D) Phaeophyceae
- E) Rhodophyta

17. MATERIAL NUMBER: 10

The structure includes.....

- A) Only sporangium
- B) Only zygote
- C) Sporangium and zygote
- D) Only egg.
- E) Egg and sperms

18. MATERIAL NUMBER: 11

Which of the followings includes this organism?

- A) Zygomycota
- B) Ascomycete
- C) Basidiomycete
- D) Phaeophyceae
- E) Rhodophyta

19. MATERIAL NUMBER: 11

The organism includes.....in its structure.

- A) Zygote
- B) Lamellate structure
- C) Basidium
- D) Gametophyte
- E) Ascus

20. MATERIAL NUMBER: 12

The dark structures on the section of this fungus are.....

- A) Spores
- B) Spore vesicles
- C) Ascus
- D) Zygote
- E) Hypha

21. MATERIAL NUMBER: 12

Which of the followings does include this fungus?

- A) Zygomycota
- B) Ascomycete
- C) Basidiomycete
- D) Rhodophyta
- E) Phaeophyceae

22. MATERIAL NUMBER: 13

This structure of the fern is.....

- A) Only antheridium
- B) Only archegonium
- C) Antheridium and archegonium
- D) Young sporophyte
- E) Strobilus

23. MATERIAL NUMBER: 14

This structure which belongs to lycopsida is.....

- A) Only antheridia
- B) Only archegonia
- C) Antheridia and Archegonia
- D) Rhizome
- E) Strobila

24. MATERIAL NUMBER: 15

This section of leaf belongs to.....

- A) Fungi
- B) A lycopside
- C) Horsetail
- D) Pine
- E) Grass

25. MATERIAL NUMBER: 15

This plant is

- A) Hydrophyte
- B) Hygrophyte
- C) Xerophyte
- D) Tropophyte
- E) Helophyte

26. MATERIAL NUMBER: 15

The big spaces at the edges are

- A) Latex
- B) Resin channels
- C) Secreting pockets
- D) Xylem tubes
- E) Phloem tubes

27. MATERIAL NUMBER: 15

Stomata occur at.....relative to the leaf surface.

- A) The same level
- B) The upper level
- C) The lower level
- D) Extremely upper
- E) There are no stomata

28. MATERIAL NUMBER: 15

The shape of parenchyma cell is

- A) Circle
- B) Star
- C) Rectangle
- D) Isodiametric
- E) Hexagonal

29. MATERIAL NUMBER: 16

The pollen belongs to.....plant.

- A) Magnolia
- B) Ranunculus
- C) Zea
- D) Pinus
- E) Solanum

30. MATERIAL NUMBER: 16

The distribution of pollen is done by.....

- A) Water
- B) Insects
- C) Wind
- D) Birds
- E) Mammalia

31. MATERIAL NUMBER: 17

This section of root belongs to.....

- A) Cocos
- B) Monstrea
- C) Orchis
- D) Ranunculus
- E) Sphagnum

32. MATERIAL NUMBER: 18

This section of root belongs to.....

- A) Astaraceae
- B) Fabaceae
- C) Araceae
- D) Lamiaceae
- E) Cactaceae

33. MATERIAL NUMBER: 19

This section of stem belongs to.....

- A) Poaceae
- B) Solanaceae
- C) Oleaceae
- D) Phaeophyceae
- E) Phagaceae

34. MATERIAL NUMBER: 20

This section of stem belongs to.....

- A) Fabaceae
- B) Pinophyta
- C) Lycopodiophyta

- D) Liliaceae
- E) Muscopsida

35. MATERIAL NUMBER: 21

This section consists of.....

- A) Only cork cambium
- B) Only lenticel
- C) Only cork tissue
- D) Only cork cambium and lenticel
- E) Cork cambium, lenticel, cork

36. MATERIAL NUMBER: 22

This section of leaf belongs to a plant.

- A) Polyticum
- B) Pinus
- C) Marchantia
- D) Triticum
- E) Quercus

37. MATERIAL NUMBER: 23

This leaf belongs to a..... plant.

- A) Hydrophytic
- B) Succulent
- C) Mesophytic
- D) Xerophytic
- E) Carnivore

38. MATERIAL NUMBER: 24

This leaf belongs to a..... plant.

- A) Hydrophytic
- B) Succulent
- C) Mesophytic
- D) Xerophytic
- E) Carnivore

39. MATERIAL NUMBER: 25

This leaf belongs to a..... plant.

- A) Hydrophytic
- B) Succulent
- C) Mesophytic
- D) Xerophytic
- E) Carnivore

40. MATERIAL NUMBER: 26

This flower belongs to a plant.

- A) Triticum
- B) Lamium
- C) Syringa
- D) Opuntia
- E) Ginkgo

41. MATERIAL NUMBER: 27

This structure which belongs to a lilium is.....

- A) Ovulary
- B) Tetrad

- C) Anther
- D) Embryo
- E) Seed

42. MATERIAL NUMBER: 28

This tissue which is seen at the outermost part of the stem is.

- A) Epidermis
- B) Cork
- C) Lenticel
- D) Stoma
- E) Trichome

43. MATERIAL NUMBER: 29

The supportive tissue in this structure is.....

- A) Sclereid
- B) Schlerenchyma
- C) Collenchyma
- D) Tracheid
- E) Trachea

44. MATERIAL NUMBER: 30

The crystal shape in this tissue is a.....

- A) Raphides
- B) Druse
- C) Prism
- D) Cubic
- E) Sand

45. MATERIAL NUMBER: 31

The pink lines between the lumens are.....

- A) Middle lamella
- B) Secondary cell wall
- C) Primary cell wall
- D) Plasmodesmata
- E) Edged passage

46. MATERIAL NUMBER: 32

Which of the followings isn't found in this root structure?

- A) Epidermis
- B) Cortex
- C) Starch
- D) Vascular bundle
- E) Cork cambium

8th INTERNATIONAL BIOLOGY OLYMPIAD

PRACTICAL TEST-B

ZOOLOGY AND ZOOLOGIC ECOLOGY

1. **MATERIAL NUMBER: 1**
This organism belongs to.....
A) Amoebae
B) Radiolarian
C) Foraminifers
D) Paramecium
E) Euspongia
2. **MATERIAL NUMBER: 1**
This organism moves using.....
A) Cilia
B) Flagellum
C) Pseudopod
D) Myosin filaments
E) Flow of water
3. **MATERIAL NUMBER: 2**
This organism belongs to.....
A) Rhodophyta
B) Phaeophyta
C) Euglenophyta
D) Chlorophyta
E) Poriferan
4. **MATERIAL NUMBER: 2**
This organism hasflagella/ flagellum.
A) 1
B) 2
C) 4
D) No
E) Many
5. **MATERIAL NUMBER: 3**
This organism is
A) Euglena
B) Trypanosome
C) Chlamydomonas
D) Chlorella
E) Vorticella
6. **MATERIAL NUMBER: 3**
The habitat of this organism is.....
A) Fresh Water
B) Salty Water
C) Intestine
D) Mouth of Mammals
E) Human's blood
7. **MATERIAL NUMBER: 3**
This organism moves using.....

- A) Cilia
- B) Flagellum
- C) Pseudopod
- D) Myosin filament
- E) Flow of water

8. **MATERIAL NUMBER: 4**

This organism is.....

- A) Euglena
- B) Paramecium
- C) Chlorella
- D) Plasmodium
- E) Vorticella

9. **MATERIAL NUMBER: 4**

This organism moves using.....

- A) Cilia
- B) Flagellum
- C) Pseudopod
- D) Myosin filament
- E) Flow of water

10. **MATERIAL NUMBER: 5**

The reproductive pattern of this single celled organism is.....

- A) Conjugation
- B) Fission
- C) Isogamy
- D) Anisogamy
- E) Autogamy

11. **MATERIAL NUMBER: 6**

This organism is.....

- A) Euglena
- B) Trypanosome
- C) Chlamydomonas
- D) Chlorella
- E) Vorticella

12. **MATERIAL NUMBER: 7**

This organism belongs to.....

- A) Hydrozoan
- B) Scyphozoan
- C) Anthozoan
- D) Euspongia
- E) Oligochaete

13. **MATERIAL NUMBER: 7**

This reproduction pattern is.....

- A) Conjugation
- B) Isogamy
- C) Budding
- D) Oogamy
- E) Fission

14. MATERIAL NUMBER: 7

The habitat of this organism is.....

- A) Salty Water
- B) Human Intestine
- C) Human Blood
- D) Fresh Water
- E) On the mosses

15. MATERIAL NUMBER: 8

This organism belongs to.....

- A) Hydrozoan
- B) Scyphozoan
- C) Anthozoan
- D) Euspongia
- E) Oligochaete

16. MATERIAL NUMBER: 9

This organisms belongs to.....

- A) Polychaete
- B) Oligochaete
- C) Hirudinea
- D) Crustacean
- E) Chelicerata

17. MATERIAL NUMBER: 9

This organism has antennae.

- A) No
- B) 2
- C) 2 pairs
- D) Brush like
- E) 4 pairs

18. MATERIAL NUMBER: 10

This organism belongs to.....

- A) Chelicerata
- B) Thysanuran
- C) Isopteran
- D) Crustacean
- E) Hymenoptera

19. MATERIAL NUMBER: 10

This organism has antennae.

- A) No
- B) 2
- C) 2 pairs
- D) Brush like
- E) 4 pairs

20. MATERIAL NUMBER: 10

This organism has legs.

- A) 4
- B) 4 pairs
- C) 3
- D) 3 pairs
- E) No

21. MATERIAL NUMBER: 10

The mouth structure is

- A) Chewer
- B) Grazer
- C) Cheliser
- D) Holder
- E) Piercer and absorbent

22. MATERIAL NUMBER: 11

..... has this kind of eye structure.

- A) Leptinotarsa
- B) Hirudo
- C) Fasciola
- D) Sepia
- E) Asteria

23. MATERIAL NUMBER: 12

This metamorphosis occurs.....

- A) Under Soil
- B) On the branches
- C) On the mammals' skin
- D) In fresh water
- E) On the death animal body

24. MATERIAL NUMBER: 12

The type of this metamorphosis is.....

- A) Ametabol
- B) Hemimetabol
- C) Holometabol
- D) Hypermetabol
- E) Neoteny

25. MATERIAL NUMBER: 12

The adult of this organism uses..... as food

- A) Larva
- B) Vegetable
- C) Fruit
- D) Mammal's blood
- E) Soil

26. MATERIAL NUMBER: 12

This organism has antennae.

- A) Many
- B) No
- C) 2
- D) 2 pairs
- E) 6

27. MATERIAL NUMBER: 13

The type of egg that show this kind of segmentation is.....

- A) Isolecithal
- B) Mesolecithal
- C) Telolecithal

- D) Centrolecithal
- E) Polylecithal

28. MATERIAL NUMBER: 13

The type of segmentation is.....

- A) Superficial
- B) Mesoblastic
- C) Meroblastic
- D) Holoblastic
- E) Discoidal

29. MATERIAL NUMBER: 13

..... show this kind of division.

- A) Frogs
- B) Birds
- C) Turtles
- D) Starfish
- E) Salamander

30. MATERIAL NUMBER: 14

This blastula stage is formed by.....type egg.

- A) Isolecithal
- B) Mesolecithal
- C) Telolecithal
- D) Centrolecithal
- E) Polylecithal

31. MATERIAL NUMBER: 14

The type of segmentation.....

- A) Superficial
- B) Mesoblastic
- C) Meroblastic
- D) Holoblastic
- E) Discoidal

32. MATERIAL NUMBER: 14

This blastula stage is formed by.....

- A) Frogs
- B) Birds
- C) Turtles
- D) Starfish
- E) Salamander

33. MATERIAL NUMBER: 15

This structure is the..... tissue of a mammal.

- A) Cartilage
- B) Bone
- C) Pancreas
- D) Thyroid
- E) Secretory

34. MATERIAL NUMBER: 15

The big gap in the middle is.....

- A) Secretory Channels
- B) Langerhans Islands

- C) Secretory Lumen
- D) Haversian Channels
- E) Chondrocytes sets

35. MATERIAL NUMBER: 16

This blood tissue belongs to.....

- A) Locusta
- B) Myotis
- C) Cebus
- D) Lumbricus
- E) Anuran

36. MATERIAL NUMBER: 17

Circular structure observed in the cross section of skin is.....

- A) Adipose tissue
- B) Vater paccini
- C) Crause
- D) Ruffini
- E) Meissner

37. MATERIAL NUMBER: 17

The duty of this structure is.....

- A) Storing lipid
- B) Sense the pressure
- C) Sensing the high temperature
- D) Sensing the low temperature
- E) Secreting sweat

38. MATERIAL NUMBER: 18

This vessel that cats have is.....

- A) Artery
- B) Vena
- C) Arteriole
- D) Lymphatic
- E) Arteriosclerosis Vena

39. MATERIAL NUMBER: 19

The type of epithelium is.....

- A) Transitional
- B) Pseudo Stratified
- C) Stratified
- D) Cubic
- E) Goblet

40. MATERIAL NUMBER: 20

This person, whose bar bodies has been stained, has.....

- A) XX
- B) XY
- C) XXX
- D) XXY
- E) XXXY

41. MATERIAL NUMBER: 21

These jaw bones belongs to.....

- A) Mammalia

- B) Piciformes
- C) Testudinate
- D) Caudata
- E) Teleostei

42. MATERIAL NUMBER: 22

This skull belongs to.....

- A) Anura
- B) Myotis
- C) Lacerta
- D) Clupea
- E) Petromyzon

43. MATERIAL NUMBER: 23

This tooth belongs to.....

- A) Strix
- B) Ciconia
- C) Syciorhinus
- D) Ascidia
- E) Dryocopus

44. MATERIAL NUMBER: 24

This human bone is.....

- A) Atlas
- B) Lumbar
- C) Thoracic
- D) Cervical
- E) Axis

45. MATERIAL NUMBER: 25

This human bone is.....

- A) Atlas
- B) Lumbar
- C) Thoracic
- D) Cervical
- E) Axis

8th INTERNATIONAL BIOLOGY OLYMPIAD
PRATICAL TEST-C
TAXONOMY

1- MATERIAL NUMBER: 1

This organism belongs to

- A. Porifera
- B. Cnidaria
- C. Turbellaria
- D. Trematoda
- E. Cestoda

2- MATERIAL NUMBER: 1

Stained part is system of that organism.

- A. Respiratory
- B. Excretory
- C. Digestive
- D. Nervous
- E. Reproductive

3- MATERIAL NUMBER: 2

This organism belongs to

- A. Porifera
- B. Cnidaria
- C. Turbellaria
- D. Trematoda
- E. Cestoda

4- MATERIAL NUMBER: 2

..... is not observed in this slide

- A. Intestine
- B. Uterus
- C. Testis
- D. Ovary
- E. Protonephridium

5- MATERIAL NUMBER: 3

This organism belongs to

- A. Platyhelminthes
- B. Nematelminthes
- C. Annelida
- D. Oligochaeta
- E. Polychaeta

6- MATERIAL NUMBER: 3

The sex of the individuals in the slide is

- A. Only male
- B. Only female
- C. Male and female
- D. Hermaphrodite

7- MATERIAL NUMBER: 4

This organism belongs to

- A. Stellozoa
- B. Echinozoa
- C. Gastropoda
- D. Scyphozoa

E. Ascidia

8- MATERIAL NUMBER: 5

This organism belongs to

- A. Turbellaria
- B. Trematoda
- C. Cestoda
- D. Hirudinea
- E. Polychaeta

9- MATERIAL NUMBER: 6

This organism belongs to

- A. Turbellaria
- B. Trematoda
- C. Cestoda
- D. Hirudinea
- E. Polychaeta

10- MATERIAL NUMBER: 6

The name of the species in this slide is

- A. Nereis
- B. Hirudo
- C. Asterias
- D. Polycelis
- E. Scolopendra

11- MATERIAL NUMBER: 7

This organism belongs to

- A. Crustacea
- B. Chelicerata
- C. Tracheata
- D. Chilopoda
- E. Homoptera

12- MATERIAL NUMBER: 7

This organism has antennae

- A. No
- B. One pair
- C. Two pairs
- D. Four pairs
- E. Six

13- MATERIAL NUMBER: 7

The name of the species in this slide is

- A. Astacus
- B. Daphnia
- C. Cyclops
- D. Aeneus
- E. Aphis

14- MATERIAL NUMBER: 8

This organism belongs to

- A. Crustacea
- B. Chelicerata
- C. Tracheata

- D. Chilopoda
- E. Hymenoptera

15- MATERIAL NUMBER: 8

This organism has antennae

- A. No
- B. One pair
- C. Two pairs
- D. Four pairs
- E. Six

16- MATERIAL NUMBER: 8

The name of the species in this slide is

- A. Ixodes
- B. Scolopendra
- C. Pediculus
- D. Leptinotarsa
- E. Gerris

17- MATERIAL NUMBER: 8

The the number of legs in this organism is

- A. Two pairs
- B. Three pairs
- C. Four pairs
- D. Six pairs
- E. Eight pairs

18- MATERIAL NUMBER: 9

This organism belongs to

- A. Diptera
- B. Hymenoptera
- C. Anoplura
- D. Isoptera
- E. Odonata

19- MATERIAL NUMBER: 9

Mouth parts are

- A. Sucker
- B. Chewer
- C. Piercing
- D. Chelicer
- E. Lapping for pollen grains

20- MATERIAL NUMBER: 9

The name of the species in this slide is

- A. Ichneumon
- B. Leptinotarsa
- C. Aphis
- D. Anopheles
- E. Pediculus

21- MATERIAL NUMBER: 10

This organism belongs to

- A. Diptera
- B. Hymenoptera

- C. Anoplura
- D. Isoptera
- E. Odonata

22- MATERIAL NUMBER: 10

Mouth parts are

- A. Piercing
- B. Chewer
- C. Licking
- D. Chelicer
- E. Lapping for pollen grains

23- MATERIAL NUMBER: 10

This the number of legs in this organism is

- A. Two pairs
- B. Three pairs
- C. Four pairs
- D. Six pairs
- E. Eight pairs

24- MATERIAL NUMBER: 10

The name of the species in this slide is

- A. Drosophila
- B. Araneus
- C. Formica
- D. Gerris
- E. Ixodes

25- MATERIAL NUMBER: 10

This is organism has antennae

- A. No
- B. One pair
- C. Two pair
- D. Four pair
- E. Six

26- MATERIAL NUMBER: 11

This organism belongs to

- A. Diptera
- B. Hymenoptera
- C. Anoplura
- D. Isoptera
- E. Odonata

27- MATERIAL NUMBER: 11

This number of legs in this organism is

- A. Two pairs
- B. Three pairs
- C. Four pairs
- D. Six pairs
- E. Eight pairs

28- MATERIAL NUMBER: 11

The name of the species in this slide is

- A. Drosophila

- B. Araneus
- C. Pediculus
- D. Gerris
- E. Ixodes

29- MATERIAL NUMBER: 11

This organism has antennae

- A. No
- B. One pair
- C. Two pair
- D. Four pair
- E. Six

30- MATERIAL NUMBER: 12

This fruit belongs to

- A. Asteraceae
- B. Solanaceae
- C. Lamiaceae
- D. Fabaceae
- E. Oleaceae

31- MATERIAL NUMBER: 13

This plant belongs to

- A. Solanaceae
- B. Magnoliaceae
- C. Rosaceae
- D. Ranunculaceae
- E. Brassicaceae

32- MATERIAL NUMBER: 13

Ovary position is

- A. Epigin
- B. Hypogin
- C. Perigin
- D. Perigineus
- E. Hypogineus

33- MATERIAL NUMBER: 14

This plant belongs to

- A. Araceae
- B. Arecaceae
- C. Poaceae
- D. Orchidaceae
- E. Liliaceae

34- MATERIAL NUMBER: 15

Ovary position is

- A. Epigin
- B. Hypogin
- C. Perigin
- D. Perigineus
- E. Hypogineus

35- MATERIAL NUMBER: 15

This plant belongs to (familia)

- A. Solanaceae
- B. Magnoliaceae
- C. Rosaceae
- D. Ranunculaceae
- E. Brassicaceae

36- MATERIAL NUMBER: 15

This plant belongs to

- A. Prunus
- B. Rosa
- C. Pulsatilla
- D. Syringa
- E. Malus

37- MATERIAL NUMBER: 16

This structure belongs to

- A. Rosaceae
- B. Fagaceae
- C. Brassicaceae
- D. Solanaceae
- E. Asteraceae

38- MATERIAL NUMBER: 16

This body part of a plant is

- A. Stem
- B. Bulb root
- C. Storage root
- D. Leaf
- E. Bud

39- MATERIAL NUMBER: 17

Leaves of the plant are

- A. Tripinnat
- B. Paripinnat
- C. Trifoliat
- D. Imparipinnat
- E. Bipinnat

40- MATERIAL NUMBER: 18

This structure belongs to

- A. Araceae
- B. Arecaceae
- C. Poaceae
- D. Orchidaceae
- E. Liliaceae

41- MATERIAL NUMBER: 19

This plant belongs to

- A. Asteraceae
- B. Arecaceae
- C. Araceae
- D. Rosaceae
- E. Fabaceae

8th INTERNATIONAL BIOLOGY OLYMPIAD
PRACTICAL PART-D
ECOLOGIC PROBLEMS

1. The amount of food that junco bird needs everyday is 10% of its weight. According to this information how much food does a Junco bird with 90 gram body weight needs everyday.

A) 90 gr. B) 45 gr. C) 9 gr. D) 10 gr. E) 18 gr.

Answer the question 2,3,4,5,6 according to the given information below. In the *Sitophilus oryzae* population density dependent egg laying number is given below.

Number of mite=128

Number of wheat seeds=200

The number of eggs that a female lay in each day=1,6

The number of eggs found in each seed=2,56

Percentage of adults getting out from eggs= ~ 70

Percentage of seeds that doesn't have any egg=5,0

Female, male ratio=1

2. Number of wheat mite for each seed?

A) 32.8 B) 1.56 C) 3.2 D) 0.64 E) 6.4

3. The total number of eggs that females lay everyday?

A) 204.8

B) 512

C) 327.6

D) 140

E) 102.4

4. The total number of eggs in the seeds is.....

A) 512

B) 327.6

C) 486.4

D) 163.8

E) 304

5. The number of seeds with egg is.....

A) 10

B) 6.4

C) 64

D) 100

E) 190

6. The total number of eggs that give rise to adults.....

A) 358

B) 229

C) 340

D) 114

E) 212

Answer the question 7,8,9,10,11,12 according to the information given below. The numbers of female and male individuals in the *Panaeus incidus* population (for each season) is given below

	Number of.....	Number of
	MALES.....	FEMALES
Winter.....	130.....	142
Spring.....	60.....	67
Summer.....	482.....	458
Fall.....	243.....	246

7. What's the maximum population size in year
A) 915
B) 913
C) 1828
D) 940
E) 518
8. The number of maximum female individuals in a year.
A) 915
B) 913
C) 458
D) 482
E) 1828
9. The season, in which the percentage of males is the highest, is
A) Winter
B) Spring
C) Summer
D) Fall
E) All seasons are same
10. What's the maximum percentage of male individuals in a year?
A) 54
B) 47.7
C) 51
D) 49
E) 50.05
11. The season in which the mortality number is the highest.....
A) Winter
B) Fall
C) Spring
D) Summer
12. The season in which the mortality percent is the highest.....
A) Winter
B) Fall
C) Spring
D) Summer

Answer the questions 13,14,15,16,17 according to information given below.

Following figure compares the terrestrial and aquatic areas.

	TERRESTRIAL AREA.....	AQUEOUS AREA
Total Area.....	$149 \times 10^6 \text{ km}^2$	$361 \times 10^6 \text{ km}^2$
Total Net Primary Production.....	$120 \times 10^9 \text{ ton/year}$	$60 \times 10^9 \text{ ton/year}$
Total Secondary Production.....	$0.8 \times 10^9 \text{ ton/year}$	$3.0 \times 10^9 \text{ ton/year}$
Total Food (Both animal and plant made).....	$1.3 \times 10^9 \text{ ton/year}$	$17.0 \times 10^9 \text{ ton/year}$

13. Total net primary product per km² on terrestrial area for a year is.....
- A) 85
 - B) 8.50×10^2
 - C) 805.369
 - D) 8.5×10^3
 - E) 8505×10^2
14. Total secondary product per km² on terrestrial area for a year is.....
- A) 53×10^2
 - B) 5.369
 - C) 5.3×10^3
 - D) 0.053
 - E) 0.53
15. Total food per km² on terrestrial area for a month is.....
- A) 7.27
 - B) 0.72×10^3
 - C) 2.7×10^2
 - D) 0.727
 - E) 0.00727
16. Total net primary product per km² on aqueous area for a month is.....
- A) 1.3×10^3
 - B) 13.850
 - C) 138
 - D) 1.3×10^2
 - E) 0.0138
17. Total secondary product per km² on aqueous area for a month is.....
- A) 6.92
 - B) 0.0069
 - C) 609.2
 - D) 60.29
 - E) 0.692
18. Total food per km² on aqueous area for a year is.....
- A) 47.091
 - B) 4.7
 - C) 47.7×10^3
 - D) 47.7×10^2
 - E) 4.7